

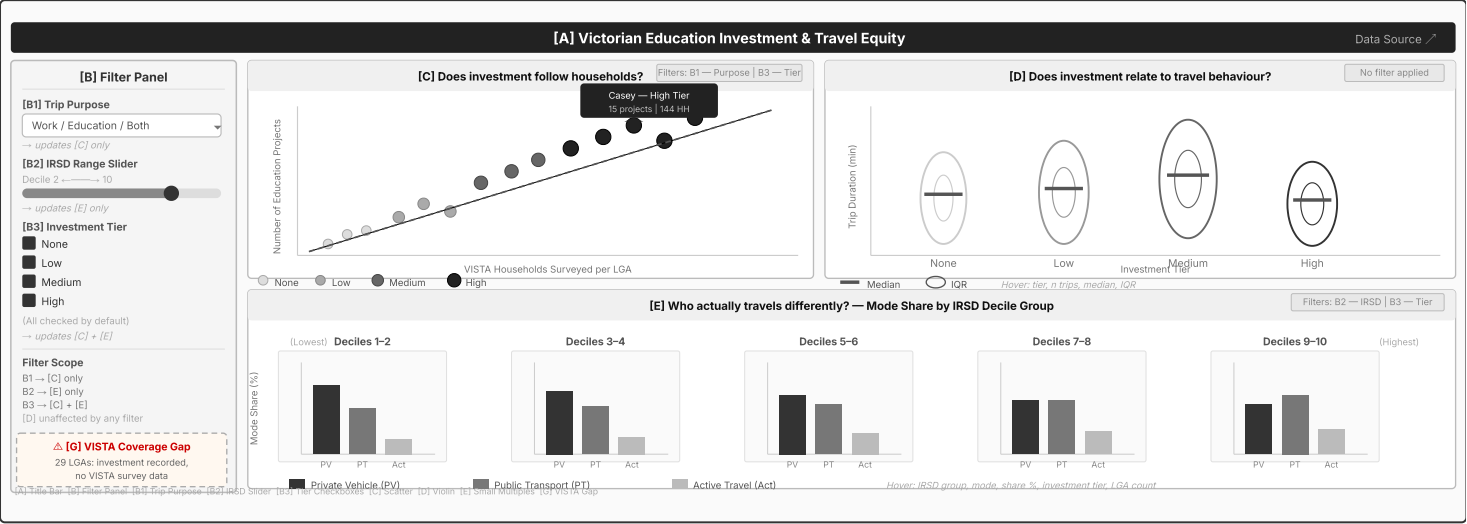
INFORMATION

Title: Design B — Question-Answer Dashboard | **Sheet No:** Sheet #3
Author: Bharath Arun Gandhimani | **Date:** 13/05/2026
Description: Fully reader-driven dashboard communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria to policymakers. Three chart panels are visible simultaneously — each poses an explicit question and answers it with a distinct encoding. A left-side filter panel scopes each filter to specific panels only.

DESIGN RATIONALE

Each panel uses a different encoding matched to its analytical task: scatter plot (bivariate investment-vs-household relationship), violin plots (distribution shape across investment tiers), and small multiples (faceted mode share by IRSD decile). Chart types drawn from Sheet 1 Ideas #5 (scatter), #4 (violin), #11 (small multiples). Filters are intentionally scoped — [B1] affects [C] only, [B2] affects [E] only, [B3] affects [C] and [E] — to prevent unintended cross-panel interference. [D] is deliberately unfiltered so the reader always has a stable distributional reference. No chart type in this design (scatter, violin, small multiples) appears in Design A or Design C — the three designs collectively cover geographic, distributional, bivariate, faceted, strip, ranking, and paired-comparison encodings across Sheets 2–4.

LAYOUT



OPERATIONS

[B1] Select Trip Purpose dropdown

[C] scatter updates to trips matching selected purpose (Work / Education / Both).
[D] and [E] unchanged — purpose-level breakdown not available at LGA-level for violin or faceted mode share.

[B2] Drag IRSD Range Slider

[E] small multiples grey out facets outside selected decile range — highlights relevant band without removing context. [C] and [D] unchanged.

[B3] Uncheck Investment Tier

[C] removes points for unchecked tier and re-draws trend line. [E] removes bars for that tier within each facet. [D] unaffected — violin shows full distribution regardless of tier.

Hover on point in [C] scatter

Tooltip: LGA name, tier, project count, household count, trip purpose. Hovered point highlighted; all others dim to 30% opacity.

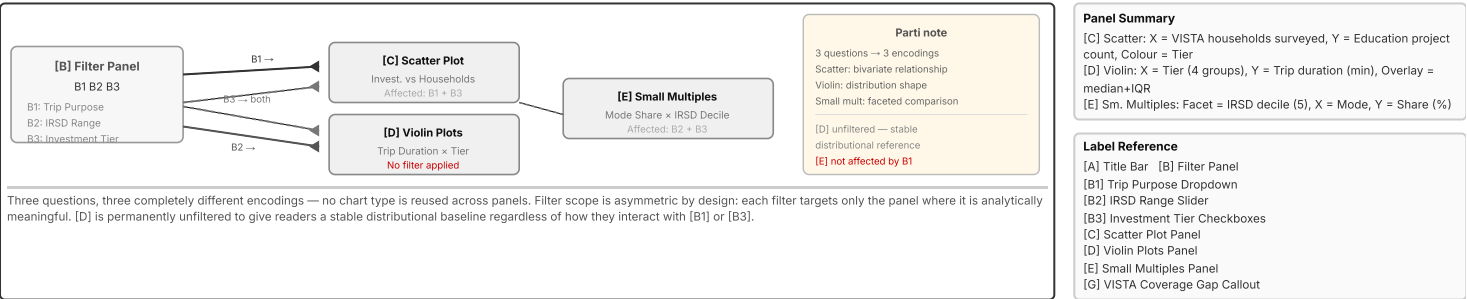
Hover on violin body in [D]

Tooltip: tier label, median duration (min), IQR range, n trips. Median and IQR markers always visible statically — hover supplements, does not replace.

Hover on bar in [E] small multiples

Tooltip: IRSD decile group, mode type, share (%), n LGAs in facet. Same mode bar highlights across all visible facets simultaneously for cross-facet comparison.

FOCUS / PARTI — FILTER COORDINATION FLOW



DISCUSSION

Pros

- A policymaker can navigate directly to the question most relevant to their brief — no compulsory sequence to sit through before reaching the panel they care about.
- Each panel uses a distinct encoding matched to its analytical task: scatter for bivariate relationship, violin for distribution shape, small multiples for faceted comparison. No chart type is repeated across panels — each question gets its own visual language.
- Asymmetric filter scoping prevents cross-panel noise — each filter only acts where it is analytically valid, reducing reader confusion from unexpected changes.
- [D] violin plots are permanently unfiltered, giving the reader a stable distributional reference that does not change as they experiment with [B1] or [B3] — useful as an anchor point.
- Small multiples in [E] let the IRSD decile gradient emerge visually across all five facets simultaneously without any interaction needed — the pattern is visible at a glance.
- R Shiny implementation is straightforward: three independent renderPlotly() outputs, each reactive only to its scoped inputs, avoiding unnecessary re-renders.

Cons

- All three panels are visible at once with no guided entry point — a policymaker unfamiliar with the topic may not know where to start and may engage only one panel, missing the full argument.
- There is no narrative thread connecting the three questions. A reader who engages only with [C] or [D] misses the socioeconomic equity dimension that [E] communicates.
- The VISTA gap callout [G] sits at the bottom of the filter panel and is easy to overlook for a reader whose attention is drawn immediately to the larger chart panels on the right.
- Asymmetric scoping (B1 → [C] only, B2 → [E] only, B3 → [C] + [E]) requires orientation time — a first-time reader may apply a filter and be confused when not all panels update.
- [C] and [E] both involve investment tier as a variable but from different angles — without accompanying explanatory text, this could be perceived as redundant or contradictory rather than complementary.
- Unlike Design A (Sheet 2), there is no closing policy implication step — the reader must draw their own conclusions from the three panels, which may be unreliable for a non-expert audience.
- [D] violin plots show trip duration across investment tiers, but the Kruskal-Wallis test performed in the Data Exploration Project found no statistically significant differences in work-trip duration across tiers — making this panel analytically weak as a centrepiece. This was a key factor in rejecting Design B in favour of Design C: a null result does not communicate a clear policy message.